

APR3044 Advanced Data Analytics - Fall 2026

[Introduction](#) | [Schedule](#) | [Policies](#)

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Class: Wed 10:00 - 13:00 in Y202-507 Lecture Room

Office hours: Thu 9:00 - 12:00 (Any time by appointment)

Prerequisites: No prior programming experience is required.
Familiarity with basic statistics, R, Python, or Excel may be helpful.

Course Description

Data analytics is increasingly important for understanding social problems, communicating evidence, and supporting data-driven decisions. As organizations and individuals work with larger and more complex forms of data, the ability to clean, explore, visualize, and communicate data has become an essential skill. At the same time, generative AI-based agents are changing how people search for information, interpret data, draft reports, automate workflows, and support decision-making processes.

This course introduces foundational concepts and practical skills in data analytics, with a focus on:

- **Data Communication:** Cleaning, exploring, visualizing, and communicating data-grounded findings.
- **AI Agent Workflows:** Applying AI tools for workflow planning, data-grounded tasks, and responsible evaluation.

Course Objectives

In this course, you will learn how to:

- Understand data structures to choose appropriate visualization methods
- Communicate data-driven findings clearly and effectively
- Design basic AI agent workflows for data analysis and communication tasks
- Implement prompt engineering, tools, and context (RAG) to build reliable AI outputs

The course combines lectures and hands-on practice. Through the final project and presentation, you will share your work, engage in discussions with peers, and develop self-directed learning and communication skills.

Course Material

- 이병관 (2023). 사회과학 연구자를 위한 R을 활용한 기초통계. 한양대학교 출판부
- Microsoft Corp. *AI Agents for Beginners - A Course* [GitHub Repository]. Retrieved from <https://github.com/microsoft/AI-Agents-For-Beginners> (Licensed under MIT)

Grades

1. Problem sets (40%)

There will be nine problem sets throughout the semester. **Your lowest problem set score will be dropped** when calculating your final problem set grade. Late submissions are not accepted. **Any assignment submitted after the deadline will receive a zero, with no exceptions.**

- (1) PS #1 (Due 9/16 11:59 am)
- (2) PS #2 (Due 9/23 11:59 am)
- (3) PS #3 (Due 9/30 11:59 am)
- (4) PS #4 (Due 10/7 11:59 am)
- (5) PS #5 (Due 10/14 11:59 am)
- (6) PS #6 (Due 10/21 11:59 am)
- (7) PS #7 (Due 11/4 11:59 am)
- (8) PS #8 (Due 11/11 11:59 am)
- (9) PS #9 (Due 11/18 11:59 am)

2. Team project (40%)

Each team will identify a real-world problem related to society, markets, or media and develop an **AI agent prototype** that uses open data to analyze and address the problem. The final project should include **data analysis & visualization**, and the agent should use these outputs to answer user questions, provide information, or support decision-making.

Example (This is only examples. You are welcome to choose any topic as long as your project uses open data and includes a AI agent prototype.

- <https://neoscholar.khu.ac.kr/hssdi/user/bbs/BMSR00040/view.do>

The team project will be evaluated through three components:

(1) Instructor Evaluation (15%)

Based on:

- i) problem definition
- ii) appropriate use of evidence and data
- iii) quality of analysis, interpretation, and recommendations
- iv) presentation skill and quality

(2) Peer Evaluation by Other Teams (5%)

Other teams will evaluate the project using the criteria provided later. A major part of this evaluation will consider how well your team responds to comments and feedback received during progress check-ins. (You are not required to incorporate every suggestion, but you should be able to provide a reasonable justification when you choose not to do so.)

(3) Peer Evaluation (15%)

Each member will evaluate the contributions of other members of their own team.

3. Participation (10%)

Participation will be evaluated through two components:

(1) Feedback to Other Teams (5%)

During project progress check-ins, each team is expected to provide comments, questions, or suggestions to other teams. Useful feedback may include points for further consideration, relevant data or information, similar cases, or constructive questions about the team's approach.

This component will be evaluated by the teams receiving the feedback. The quality and usefulness of the feedback provided will be considered.

(2) Overall Class Participation (5%)

This component will be evaluated by the instructor based on your overall engagement in class, including participation in discussions, team activities, and project-related activities.

Remember that the purpose of the team project is not necessarily to create something that has never existed before. Identifying relevant examples, cases, or data can also provide valuable feedback. The most important thing is **to listen carefully when other teams present** and contribute meaningfully when appropriate.

4. Attendance (10%)

Up to two unexcused absences are allowed without penalty. However, beginning with the third unexcused absence, 3 points will be deducted from the attendance grade for each additional absence.

Schedule

Week	Date	Title	Activity		
			Take-home	In-class	Consultation
1	2026.9.2	Orientation & Introduction to R I			
2	2026.9.9	Introduction to R II & Data Preprocessing	Problem set #1	(Team building)	
3	2026.9.16	Exploratory Data Analysis	Problem set #2	Team PT	
4	2026.9.23	Choosing the Right Chart and Building Visualizations in R	Problem set #3	Progress check-in 1	
5	2026.9.30	Interactive Data Visualization I: Tableau	Problem set #4		
6	2026.10.7	Interactive Data Visualization II: Plotly	Problem set #5		
7	2026.10.14	Wrap-up Session I: Data Visualization Workshop	Problem set #6		Team meeting day 1
8	2026.10.21	Introduction to AI Agents			
9	2026.10.28	Designing AI Agent Workflows	Problem set #7		
10	2026.11.4	Prompting, Tools, and Data-Grounded Responses	Problem set #8		
11	2026.11.11	Agentic RAG and Context Engineering	Problem set #9	Progress check-in 2	
12	2026.11.18	Planning, Memory, and Multi-Agent Collaboration		Progress check-in 3	
13	2026.11.25	Responsible and Trustworthy AI Agents		Progress check-in 4	
14	2026.12.2	Wrap-up Session II: AI Agent Project Workshop			Team meeting day 2
15	2026.12.9	Final Presentation	Course feedback	Final presentation	
16	2026.12.16	Course Wrap-up	Peer evaluation		

The course schedule is subject to change based on class progress and discussion. Any changes will be announced in advance.

Course Policies

1. AI Policy

You are welcome to use generative AI tools in this class. However, any use of AI in submitted work must be clearly acknowledged and properly documented. You are responsible for verifying the accuracy, originality, and appropriate use of AI-generated content.

For each assignment, please include a brief “AI Use Disclosure” indicating:

- Generative AI tool and model used:
 - ChatGPT (GPT-5)
 - Claude (Claude Sonnet 4)
 - Gemini (Gemini 2.5 Pro)
 - Other: _____
- Purpose(s) of use:
 - Idea generation and exploration
 - Copyediting, language improvement, or translation
 - LLM-assisted information search
 - Literature search or classification
 - Coding assistance or debugging
 - Data analysis assistance
 - Other: _____

2. Academic Integrity Policy

I have a responsibility to uphold the university’s academic integrity policy, and I take that responsibility very seriously. If you are unsure whether a particular form of assistance is acceptable, please ask me before submitting your work.

Discussion of concepts with classmates and the use of external resources or AI tools for learning are permitted. However, copying or submitting work produced by another person, source, or AI as your own is not permitted. This course maintains a **zero-tolerance policy toward cheating and plagiarism. If you cannot explain your answer, reproduce the reasoning, or complete a similar task independently, the submitted work will receive a grade of zero.** After submission, claiming that you “did not know” or were unaware of the policy will not exempt you from responsibility for an academic integrity violation.

3. Late Policy

You are expected to arrive on time. Arrival more than 15 minutes after the scheduled start of class will be recorded as late. Three late arrivals will be counted as one unexcused absence.